



SAFESTEP

Time & Crowd-Aware Women's Safe Navigation System

CODEX 2026

Specification & Formula Document
Version 2.5 Hardened • Production Ready

Executive Overview: SafeStep is an intelligent pedestrian safety platform engineered to overcome the fundamental flaw of legacy navigation algorithms (such as Google Maps), which blindly optimize for vehicle transit speed while routing solo pedestrians through unlit, isolated, and hazardous corridors. SafeStep integrates municipal road geometry, live illumination coverage, verified 24/7 sanctuaries, cryptographic tamper-proof tokens, and real-time guardian telemetry.

☀️ 1. Comprehensive Feature Architecture

A. Intelligent Navigation & Road Routing

🟢 Multi-Route Comparative Scoring

Simultaneously calculates and displays the **Safest Route** (max lighting & sanctuaries), **Fastest/Unsafe Shortcut** (warns of low visibility and zero havens), and **Standard Route**.

🗺️ 100% Real-Road Guarantee

Routes are strictly snapped to municipal walking and vehicle graphs (OSRM) with lateral road via-points. Zero linear cuts through buildings or unnavigable terrain.

🚶 Turn-by-Turn Navigation HUD

High-contrast top card showing upcoming maneuvers, current street names, distance countdowns, and directional vector icons (↩️, ➡️, ⬆️, 🚧).

🔊 Hands-Free Voice Guidance

Native Web Speech API audio synthesis speaks maneuvers and sanctuary proximity aloud, allowing users to keep their eyes up and phones secure at night.

💡 Night Illumination Underlay

Visual cyan/green luminous glow along verified lit municipal streets contrasted against dashed caution lines for dark stretches.

🗺️ Offline Manhattan Grid Routing

When external OSRM servers or 4G data drop, an internal algorithmic Manhattan grid generator produces >20 coordinates and 4–5 realistic turn maneuvers.

B. Emergency & Tactical Panic Defense Suite

🚶 One-Click Safe Haven Sprint

Instantly locks origin to the user's current GPS position and plots the shortest, direct sprint path to the closest 24/7 police post, hospital, or open pharmacy with zero detours.

📢 High-Decibel Loud Siren

Web Audio API dual-oscillator generating a piercing emergency alarm through the device speaker to startle attackers and summon civilian assistance.

📞 Simulated Fake Incoming Call

Renders an authentic incoming phone call UI with customizable caller ID ("*Inspector Vikram - Dad*") and synthesized dialogue to deter stalkers without confrontation.

📞 Direct 112 Emergency Hotline

Hardware-bridged touch shortcut directly dialing National Emergency Support (112) with pre-populated geographic context.

C. Guardian Sentinel & Live Telemetry

Guardian Status Ribbon

Pinned real-time monitoring ribbon tracking battery percentage, check-in deadlines, active trusted guardians, and live GPS accuracy.

Cryptographic Trip Rooms

Generates unguessable 128-bit URLs (/guardian/:tripToken) for real-time guardian tracking over WebSockets without exposing identity or phone numbers.

Automated Check-In Deadline

Dynamic arrival countdown timer. If the user fails to check in before the timer expires, an automated distress alert is broadcast to guardians.

Battery Sentinel Alert

Monitors device power level via Battery Status API; broadcasts a low-power snapshot with ETA and corridor position before the phone shuts off.

D. Smart City Crowdsourcing & Infrastructure Heatmap

Geo-Spatial Hazard Reporting

Enables citizens to report broken streetlights, unlit dark alleys, catcalling/harassment incidents, or physical obstacles.

Algorithmic Corroboration Engine

Single reports remain pending; multi-user independent reports within 250m dynamically corroborate and penalize corridor safety scores.

Municipal Police/PWD Intelligence

Aggregates anonymized, fuzzed incident data into infrastructure heatmaps for city authorities to prioritize street repairs and police patrols.

Anti-Defamation & Doxxing Defense

Regex-based filter blocks targeting named individuals or residential addresses while permitting genuine reports of predatory behavior.



2. Technology Stack

DOMAIN	TECHNOLOGIES & MODULES	ARCHITECTURE & ROLE
Frontend UI / UX	EJSHTML5Semantic	Server-rendered templates with clean declarative separation of concerns and zero inline JavaScript blocks.
Styling & Responsive	Custom CSS3Glassmorphism	Mobile-first responsive layout, 4×2 emergency touch grid, touch scroll ribbons, and WCAG reduced-motion compliance.
Spatial & GIS Mapping	Leaflet.js 1.9.4OpenStreetMap	Interactive vector mapping, custom SVG marker beacons, dynamic illumination corridor underlays, and polygon rendering.
Real-Time Telemetry	Socket.IO 4.8.3 (Client & Server)	Bi-directional WebSocket channels for live guardian tracking, SOS broadcast alerts, and instant crowdsourced sync.
Speech & Audio Engine	Web Speech APIWeb Audio API	Native speechSynthesis for hands-free audio turn directions and dual-oscillator high-decibel alarm generation.
Backend & API Server	Node.js 18+Express.js 5.2.1	High-throughput asynchronous REST endpoints, reverse geocode proxying, and route calculation engine.
Data Persistence	MongoDBIn-MemoryMongoose 9.10	Dual-mode persistence: MongoDB primary store + resilient in-memory fallback for zero-downtime offline execution.
GIS Services	OSRM APINominatim OSM	Open Source Routing Machine walking & vehicle road graphs + reverse geocoding with bounded memory LRU caching.
Application Security	Helmet.js 8.3Node.js Crypto	Strict CSP (zero 'unsafe-inline'), 128-bit custom HMAC-SHA256 bearer tokens, sliding-window rate limiters.
Test & QA Suite	node:assertnode:http	Comprehensive 23-test suite (17 security unit tests, 5 live HTTP integration tests, 1 offline fallback routing test).

3. Mathematical Formulations & Algorithms

1. Dynamic Composite Route Safety Score Equation

Calculated server-side in `index.js` for every candidate road path to produce an objective safety rating between 18 and 96:

$$S = \text{clamp}(18, 96, \text{round}(S_0 + \Delta S_{\text{police}} + \Delta S_{\text{haven}} + \Delta S_{\text{lighting}} - \Delta S_{\text{hazards}}))$$

- Baseline Departure Score (S_0):**

$$S_0 = 86 \text{ (if } 06:00 \leq \text{hour} \leq 19:59 \text{ [Daylight])}$$

$$S_0 = 68 \text{ (if hour} \geq 20:00 \text{ or hour} \leq 05:59 \text{ [Night])}$$

- Police Proximity Modifier (ΔS_{police}):** Evaluates Euclidean-snapped distance to the nearest police post (d_{police}):

$$\Delta S_{\text{police}} = +12 \text{ (if } d_{\text{police}} \leq 380 \text{ meters)}$$

$$\Delta S_{\text{police}} = +6 \text{ (if } 380 \text{ m} < d_{\text{police}} \leq 750 \text{ meters)}$$

$$\Delta S_{\text{police}} = 0 \text{ (if } d_{\text{police}} > 750 \text{ meters)}$$

- Commercial & Medical Sanctuary Modifier (ΔS_{haven}):** Evaluates open 24/7 facilities within 300 meters:

$$\Delta S_{\text{haven}} = (\text{has24x7Pharmacy} \times 6) + \min(10, N_{\text{medical_havens}} \times 5)$$

- Municipal Lighting Illumination Modifier ($\Delta S_{\text{lighting}}$):** Based on recorded lighting percentage ($L \in [0, 100]$):

$$\Delta S_{\text{lighting}} = +10 \text{ (if } L \geq 75\%)$$

$$\Delta S_{\text{lighting}} = -16 \text{ (if } L < 45\% \text{ and isNight)}$$

$$\Delta S_{\text{lighting}} = 0 \text{ (otherwise)}$$

- Corroborated Hazard Penalty ($\Delta S_{\text{hazards}}$):** Active hazards within 250 meters of route corridor:

$$\Delta S_{\text{hazards}} = 22 \times N_{\text{hazards}}$$

- Safety Rating Badge Classification:**

$$\text{Badge} = \text{🟢 High Safety } (S \geq 85)$$

$$\text{Badge} = \text{🟡 Moderate } (60 \leq S < 85)$$

$$\text{Badge} = \text{🚨 Low Safety } (S < 60)$$

2. Haversine Great-Circle Geodesic Distance Formula

Calculates the true curved-earth walking distance in meters between two coordinates ($\text{lat}_1, \text{lng}_1$) and ($\text{lat}_2, \text{lng}_2$):

$$\Delta\phi = (\text{lat}_2 - \text{lat}_1) \times (\pi / 180)$$

$$\Delta\lambda = (\text{lng}_2 - \text{lng}_1) \times (\pi / 180)$$

$$a = \sin^2(\Delta\phi / 2) + \cos(\text{lat}_1 \times \pi/180) \times \cos(\text{lat}_2 \times \pi/180) \times \sin^2(\Delta\lambda / 2)$$

$$c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1 - a})$$

$$\text{Distance (m)} = R \times c \text{ (where } R = 6,371,000 \text{ meters)}$$

3. Safe Haven Weighted Priority Distance Formulation

When 🏠 **Safe Haven** is activated, SafeStep computes weighted response distance prioritizing fortified 24/7 facilities:

$$D_{\text{weighted}} = D_{\text{actual}} \times W_{\text{facility}}$$

$$\text{Selected Sanctuary} = \text{argmin}_i (D_{\text{actual}_i} \times W_{\text{facility}_i})$$

- **Police Station / Post:** $W = 0.75$ (25% virtual distance preference)
- **24/7 Open Pharmacy:** $W = 0.85$ (15% virtual distance preference)
- **Hospital / Emergency Facility:** $W = 0.90$ (10% virtual distance preference)
- **General Safe Haven:** $W = 1.00$

4. Privacy-Preserving Gaussian Spatial Fuzzing Formula

Crowdsourced incident report coordinates undergo spatial obfuscation before persistence to prevent stalker reverse-triangulation:

$$\text{lat_fuzzed} = \text{round}(\text{lat} + N(0, \sigma / 111,320), 3)$$

$$\text{lng_fuzzed} = \text{round}(\text{lng} + N(0, \sigma / (111,320 \times \cos(\text{lat}))), 3)$$

Where $\sigma \approx 50$ meters, and rounding to 3 decimal places enforces a privacy protection perimeter of ± 110 meters.